CSE 4345: Big Data Analytics

Week 5, Part 1 — Handling Big Data Files: YARN, File Formats & Optimization

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	5 of 11 | Part 1 of 2
CLO:	CLO3
PLO:	PLO(e), PLO(f)

CLO3

"Explain big data techniques, tools, and technologies used to store, manage, and analyse large datasets"

Course Progression

Week 3: HDFS (where data lives)
Week 4: MapReduce / Pig / Hive / HBase (how

Part 1 Covers

- Why storage format and scheduling matter at scale
- Hadoop 1.x vs. Hadoop 2.x: the YARN revolution
- YARN three-component architecture
- Container lifecycle and job execution flow
- File formats: CSV/Text, Avro, Parquet, ORC
- Row-oriented vs. columnar storage (why it matters)
- Compression codecs: Snappy, Gzip, LZO, Zstd
- Splittability: the critical property
- Format selection matrix for production
- Ivy League alignment

Lecture Agenda

The Storage & Scheduling Problem

Why Format and Scheduling Matter

The Two Questions After HDFS & MapReduce

Once we can *store* data (HDFS) and *compute* over it (MapReduce), two engineering questions dominate production performance:

1. **Scheduling:** How do we share a 1,000-node cluster across 50 simultaneous jobs, different teams, and different frameworks (MapReduce, Spark, Tez) — fairly and efficiently?
2. **Format:** How do we store data so that a query reading 3 of 100 columns reads 3% of the data, not 100%?

Real Cost at Scale

A 100-column table with 10 TB of data. A query selects 3

Hadoop 1.x Bottleneck

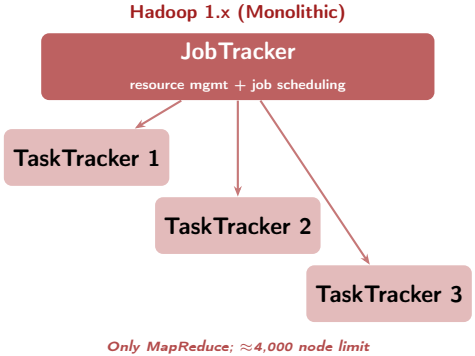
In Hadoop 1.x, MapReduce v1 bundled resource management with computation. Every cluster resource was managed by the **JobTracker**, which:

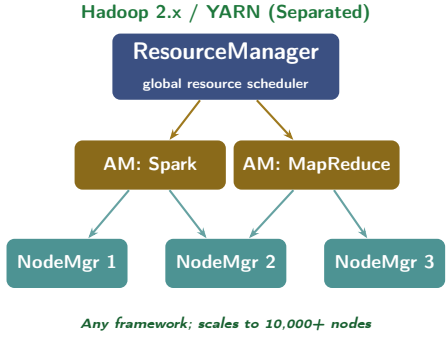
- Only ran MapReduce jobs (Spark, Tez, Storm had no path to cluster resources)
- Had a single JobTracker as a bottleneck ($\approx$4,000 node limit)
- Mixed cluster scheduling with job coordination

The fix: Separate resource management from computation. This became **YARN** in Hadoop 2.0 (2013).

YARN: The Resource Management Revolution

Hadoop 1 vs. Hadoop 2: The YARN Split





The Key Architectural Decision

YARN introduced the **ApplicationMaster** — a per-job component that *negotiates* resources from the ResourceManager. This decoupling means any framework (Spark, Flink, Tez) can run on a YARN cluster without knowing how the cluster scheduler works.

YARN Three-Component Architecture

1. ResourceManager (RM)

Global cluster resource scheduler. Runs on a dedicated master node.

- **Scheduler**: allocates *containers* to ApplicationMasters; pluggable (FIFO, Fair, Capacity)
- **ApplicationsManager**: accepts job submissions; starts the first container for each ApplicationMaster

2. NodeManager (NM)

Per-node agent running on every worker node.

- Launches and monitors **containers** on the local machine

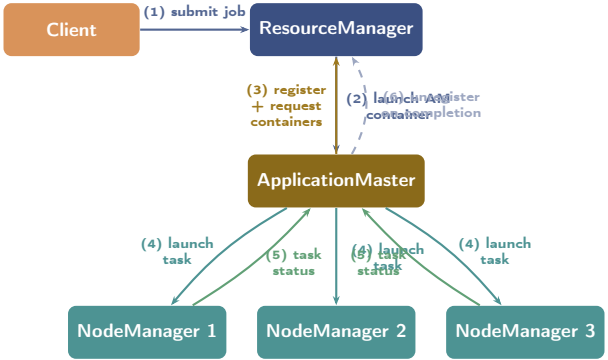
3. ApplicationMaster (AM)

One per running job — the job's own scheduler.

- Requests resources (containers) from the RM on behalf of the job
- Tells the NM to launch specific tasks inside those containers
- Monitors task progress; re-requests containers for failed tasks
- Releases containers when tasks complete

Example: A Spark AM requests 100 containers (each 4 cores, 8 GB RAM) for Spark Executors; the RM grants 80 (cluster is busy); the AM assigns tasks to those 80 Executors.

YARN Job Execution Flow



YARN Fault Tolerance

If a **NodeManager fails**, the RM detects via missed heartbeats and marks all containers on that node as failed; the AM re-requests replacement containers. If the **AM itself fails**, the RM can restart a new AM (configured via `yarn.resourcemanager.am.max-attempts`).

YARN Schedulers: Sharing the Cluster

Three Built-in Scheduler Policies

1. **FIFO Scheduler**: jobs run in submission order; simple but not suitable for multi-tenant clusters (one long job starves all others)
2. **Capacity Scheduler** (default in HDP/CDH): cluster divided into queues with guaranteed capacity; e.g., Engineering queue = 60%, Finance queue = 40%; elastically borrows unused capacity
3. **Fair Scheduler** (default in Cloudera): all jobs get equal share of resources; latency-sensitive short jobs complete quickly even if a large job is running

Bangladesh Context: a2i Cluster Sharing

The a2i (Access to Information) programme runs analytics on government data:

- **Queue A (60%)**: nightly DGHS health-record batch jobs (MapReduce)
- **Queue B (30%)**: Spark streaming for real-time agriculture data from DAE
- **Queue C (10%)**: ad-hoc Hive queries by data analysts

With **Capacity Scheduler**, queue B cannot monopolise resources even during peak streaming periods; queue A is guaranteed its 60%.

DRF: Dominant Resource Fairness

File Formats: Row vs. Columnar Storage

The Row vs. Columnar Storage Fundamental

Row Store (CSV / JSON)

id	name	age	score
1	Alice	24	91
2	Bob	31	78
3	Carol	27	88
4	Dave	29	95

Query reads ALL columns

Columnar Store (Parquet / ORC)

id	name	age	score
1	Alice	24	91
2	Bob	31	78
3	Carol	27	88
4	Dave	29	95

Query reads ONLY score column

I/O Reduction at Scale

Row store: `SELECT score FROM users` reads ALL columns (id, name, age, score). **Columnar store**: reads ONLY the score column — physically skipping id, name, age on disk. For a 100-column table, this is a **97% I/O reduction** for a 3-column query.

File Format Deep Dive: Avro, Parquet, ORC

Format	Orientation	Schema	Splittable	Compression	Best For
Text / CSV	Row	External	With codec	None (default)	Portability, small data
JSON (lines)	Row	Self	No (raw)	External	APIs, log ingestion
Avro	Row	Self-describing (JSON)	Yes	Snappy/Deflate	Schema evolution, Kafka → HDFS
Parquet	Columnar	Self-describing	Yes	Snappy/Gzip	Spark, Presto, ML pipelines
ORC	Columnar	Self-describing	Yes	Zlib/Snappy	Hive, highest compression

Why Avro for Schema Evolution?

Why Parquet for Analytics?

Compression Codecs & Splittability

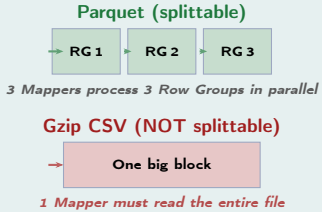
Compression Codecs: The Trade-off Landscape

Key Compression Codecs in Hadoop

Codec	Speed	Ratio	Splittable?
Snappy	Fast	Medium	No (but Parquet row groups are)
Gzip	Slow	High	No
LZO	Fast	Medium	Yes (with index)
Bzip2	Slowest	Highest	Yes
Zstd	Fast	High	No (block-level)

What Does "Splittable" Mean?

A file format is **splittable** if HDFS can start reading from any point in the file without needing to decompress from the beginning.

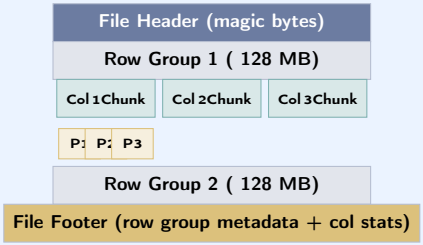


Best Practice

Use **Parquet + Snappy** for analytics (Spark, Hive).
Use **Avro + Snappy** for streaming ingest (Kafka →

Inside Parquet: Row Groups, Column Chunks, Pages

Parquet Internal Layout



Why This Layout Enables Predicate Pushdown

Query: `WHERE score > 90`

1. Read **File Footer** only: get column stats (min/max) per row group per column chunk
2. Row Group 1 has score in [60, 85] $\rightarrow$ **skip entire row group** (no I/O)
3. Row Group 2 has score in [85, 99] $\rightarrow$ must read this chunk
4. Within Row Group 2: read only the **score column chunk**

$\Rightarrow$ A query on 10 TB Parquet may physically read only **50 GB**.

Format Selection in Production

Format Selection Decision Matrix

Scenario	Recommended Format	Compression	Reason
Streaming ingest (Kafka → HDFS)	Avro	Snappy	Schema evolution; row-append friendly
Spark analytics pipeline	Parquet	Snappy	Columnar; predicate pushdown; Spark native
Hive SQL warehouse	ORC	Zlib	Best compression; Hive ACID support; vectorized reads
Data exchange / APIs	Avro or JSON	None	Human-readable schema; broad ecosystem support

Ivy League Alignment

YARN & File Formats in Top University Curricula

Harvard CS265 — Big Data Systems (Idreos)

Harvard frames file formats as a **fundamental algorithm design decision**:

- **I/O complexity model**: the cost of an algorithm is measured in the number of *I/O operations* (not time), and columnar formats directly reduce this measure
- **Access path selection**: just as a database optimizer chooses between a full table scan and an index, Parquet's predicate pushdown acts as a *lightweight index* embedded in the footer
- Harvard's benchmark: Parquet with predicate pushdown reduces I/O by 10–100$\times$ for selective queries on wide tables (100+ columns)

Harvard exam question: *"Prove that for a query selecting k columns from a c-column row-store table of n rows, the I/O complexity is $\Theta(n \cdot c)$, but $\Theta(n \cdot k)$ for a column store."*

MIT 6.830 — Database Systems

MIT compares YARN's DRF scheduler with classical database concurrency control:

- **DRF** (Ghodsi et al., 2011): generalises max-min fairness to multiple resource dimensions — the dominant resource of a job determines its fair share
- The YARN Capacity Scheduler is a practical implementation of hierarchical DRF with preemption

CMU 15-721 — Advanced DB Systems

CMU covers Parquet/ORC as modern implementations of the **DSM (Decomposition**

Student Assessment Pool

Assessment—MCQ Set A

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Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. Which component in YARN is responsible for **global resource allocation** across the entire cluster?

- NodeManager (b) ApplicationMaster (c) **ResourceManager** (d) NameNode

Q2. Which Hadoop file format stores data in **columnar** format and is best suited for analytical, read-heavy workloads?

- Avro (b) Text/CSV (c) **Parquet** (d) JSON

Instructor Notes

Q1: Students confuse ApplicationMaster (per-job) with ResourceManager (global). Reinforce: **one RM per cluster**, one AM per running application. **Q2:** Students may choose ORC (also correct for Hive, but Parquet is the *broader* standard for analytics across Spark, Presto, and ML). Accept ORC as partially correct with justification.

Assessment—MCQ Set B

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Q3. What does **"splittable"** mean for a file format in Hadoop?

a. The file can be deleted in parts by HDFS
b. **Multiple Mappers can process different parts of the file simultaneously**
c. The file format supports splitting schema across multiple NameNodes
d. The file can be distributed across multiple data centres

Q4. YARN was introduced in which major Hadoop version, and what did it replace?

a. Hadoop 1.0; replaced the NameNode
b. **Hadoop 2.0; replaced the monolithic JobTracker with a decoupled ResourceManager + ApplicationMaster**
c. Hadoop 3.0; replaced DataNodes with Containers
d. Hadoop 2.0; replaced the Secondary NameNode with a Standby NameNode

Instructor Notes

Q3: Option (b) is the correct answer. Splittability enables data-local task scheduling. **Q4:** Option (d) is a common distractor (HA NameNode is also a Hadoop 2.0 feature, but is unrelated to YARN). Students must know the JobTracker $\rightarrow$ RM + AM split.

Assessment—True / False

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Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	Answer
1. Parquet is a row-oriented file format optimised for write-heavy workloads.	**False**
2. YARN allows multiple computation frameworks such as Spark and Tez to share resources on the same Hadoop cluster.	**True**
3. Gzip-compressed files are natively splittable in Hadoop, allowing multiple Mappers to process them in parallel.	**False**
4. ORC format is specifically optimised for Hive workloads and achieves among the highest compression ratios of any Hadoop format.	**True**

Assessment—Descriptive Questions *CLO3*

Instructions

Answer each question in **100–150 words**. Include a diagram where indicated.

D1. Explain YARN's **three-component architecture** — ResourceManager, NodeManager, and ApplicationMaster. Draw a diagram showing how they interact when a Spark job is submitted to the cluster.

D2. What is **columnar storage**? Why does it improve performance for analytics queries that select only a subset of columns? Use a concrete example with numbers to illustrate the I/O reduction.

D3. Compare **Avro** and **Parquet**. Explain what "schema evolution" means and why Avro handles it better. In which scenarios would you use each?

Assessment—Analytical Questions *CLO3*

Instructions

Answer in **200–300 words** with step-by-step reasoning. Marks awarded for correct process, not just final answer.

A1. Data Lake Performance Migration:

A Dhaka-based telecom company stores 10 TB of daily call-detail records (CDR) on HDFS as gzip-compressed CSV files. Their data engineers report that a Hive query selecting 4 of 80 columns takes 6 hours.

- Identify **two specific reasons** why this configuration is slow
- Propose a migration strategy: new format, compression codec, and partitioning scheme
- Estimate the expected I/O reduction (show your calculation)
- What YARN scheduler would you use and why, given that multiple teams share the cluster?

A2. YARN Failure Trace:

A NodeManager (NM2) hosting 5 running containers for a MapReduce job fails mid-execution. Trace the complete failure recovery sequence: which components detect the failure, in what order, what state changes occur in the ResourceManager, and how are the 5 containers' tasks restarted.

Textbook & Reading References

Primary Textbooks for Week 5

- **[TW]** Tom White. *Hadoop: The Definitive Guide*, O'Reilly, 4th ed., 2015. **Ch. 3, 4, 5**
  - Ch. 3: HDFS · Ch. 4: YARN · Ch. 5: Hadoop I/O (compression, serialization, file formats)
- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 6**
  - Ch. 6: YARN and Resource Management

Supplementary (Covered in Week 5, Part 2 — Seminal Paper & Case Study)

- Vavilapalli, V.K. et al. (2013). **Apache Hadoop YARN: Yet Another Resource Negotiator.** *ACM SoCC 2013*. **[PRIMARY SEMINAL PAPER]**
- Ghodsi, A. et al. (2011). **Dominant Resource Fairness: Fair Allocation of Multiple Resource Types.** *USENIX NSDI 2011*. **[DRF algorithm underlying YARN Fair Scheduler]**
- Apache Parquet Documentation: `parquet.apache.org` — Row groups, column chunks, predicate pushdown
- Apache Avro Documentation: `avro.apache.org` — Schema evolution, reader/writer schema

Questions & Discussion

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"Storing data in HDFS is cheap. Storing it in the wrong format is expensive — you pay the cost every single time you query it."

— Julien Le Dem, Apache Parquet co-creator (Twitter, 2014)

Next Lecture: Week 5, Part 2

Seminal Paper: Vavilapalli et al. (2013) — *"Apache Hadoop YARN: Yet Another Resource Negotiator"*

Design decisions · Capacity Scheduler internals · DRF algorithm

Case Study: LinkedIn's schema evolution challenge and Avro adoption (5 TB/day scale)

Reading before Part 2: [TW] Ch. 4 · YARN paper (see references)